

Empirical treatment of neonatal sepsis: are the current guidelines adequate?

B Muller-Pebody,¹ A P Johnson,¹ P T Heath,² R E Gilbert,³ K L Henderson,¹ M Sharland,⁴ iCAP Group (Improving Antibiotic Prescribing in Primary Care)

¹Department of Healthcare-Associated Infection and Antimicrobial Resistance, Health Protection Agency Centre for Infections, London, UK

²Division of Child Health, St George's, University of London, London, UK

³MRC Centre of Epidemiology for Child Health, University College London – Institute of Child Health, London, UK

⁴Paediatric Infectious Disease Unit, St George's Hospital, University of London, London, UK

Correspondence to

B Muller-Pebody, Department of Healthcare-Associated Infection and Antimicrobial Resistance, Health Protection Agency Centre for Infections, 61 Colindale Avenue, London NW9 5EQ, UK; Berit.Muller-Pebody@hpa.org

Accepted 3 April 2010

Published Online First
28 June 2010

ABSTRACT

Objectives To use national laboratory surveillance data to determine whether pathogens responsible for neonatal bacteraemia were sensitive to nationally recommended antibiotic regimens.

Design All reports of neonatal bacteraemia received by the Health Protection Agency's voluntary surveillance scheme in England and Wales from January 2006 until March 2008, were extracted from the database. Organisms were ranked by frequency, and proportions susceptible to antimicrobials recommended for empirical treatment of neonatal sepsis were determined.

Results There were 1516 reports of bacteraemia for neonates <48 h old (early-onset) and 3482 reports for neonates 2–28 days old (late-onset). For early-onset bacteraemia, group B streptococcus (GBS) was the most frequent pathogen (31%) followed by coagulase-negative staphylococci (CoNS; 22%), non-pyogenic streptococci (9%) and *Escherichia coli* (9%). For late-onset bacteraemia, CoNS were isolated most frequently (45%), followed by *Staphylococcus aureus* (13%), Enterobacteriaceae (9%), *E coli* (7%) and GBS (7%). More than 94% of organisms (early-onset) were susceptible to regimens involving combinations of penicillin with either gentamicin or amoxicillin, amoxicillin combined with cefotaxime or cefotaxime monotherapy. More than 95% of organisms (late-onset) were susceptible to gentamicin with either flucloxacillin or amoxicillin and amoxicillin with cefotaxime, but only 79% were susceptible to cefotaxime monotherapy.

Conclusions Current guidelines for empirical therapy in neonates with sepsis are appropriate. However, gentamicin-based regimens should be used in preference to cefotaxime-based treatments, because of lower levels of susceptibility to cefotaxime and the need to avoid exerting selective pressure for resistance. Surveillance data linked to clinical data should further inform rational antibiotic prescribing in neonatal units.

INTRODUCTION

Neonatal septicaemia remains a major cause of mortality and morbidity despite advances in perinatal and neonatal care.¹ In neonates, early warning symptoms of septicaemia are often minimal, but the clinical course may be fulminant. Therefore, antimicrobial treatment of neonates with suspected sepsis must start without delay. As microbiological and antimicrobial susceptibility results are not immediately available, initial antimicrobial treatment is usually empirical with the aim of being effective against the most likely pathogens. In order to guide empirical prescribing,

What is already known on this topic

- ▶ Knowledge of both the common pathogens causing septicaemia in neonates and their antimicrobial susceptibility is essential in order to select appropriate antimicrobial treatment.
- ▶ Empirical antimicrobial treatment is increasingly compromised by antimicrobial resistance.

What this study adds

- ▶ The current pattern of bacteria isolated from blood cultures taken from neonates in England and Wales is described.
- ▶ National guidelines for empirical antimicrobial therapy in neonates with sepsis appear appropriate.
- ▶ However, cefotaxime-based treatments should not be recommended, because of lower susceptibility levels and the need to avoid exerting selective pressure.

it is crucial to monitor changes in the pattern of causative organisms and their antimicrobial susceptibility profiles.

The main routes of infection for a neonate are perinatal vertical transmission and postnatal exposure to organisms usually associated with nosocomial infections. Perinatal vertical transmission manifests itself in early-onset infection during the first 2 days after birth, the most common pathogen being group B streptococcus (GBS). Following the introduction of universal screening for maternal GBS colonisation and adoption of widespread intrapartum antibiotic prophylaxis (IAP) in the USA in the mid-1990s, changes in the distribution of pathogens from predominantly gram-positive organisms to gram-negative Enterobacteriaceae, particularly *Escherichia coli*, were reported.^{2–5} However, since many factors impact on the maternal vaginal flora (age, ethnic group, underlying medical condition, community and intrapartum antibiotic treatment), the reason for this shift in causative organisms is uncertain. Recent studies in the UK report national incidence rates of GBS infection in infants similar to those

seen in the USA despite the lack of screening for antenatal GBS carriage and routine IAP use.⁶

In contrast, late-onset infections in neonates are usually nosocomial and become clinically evident more than 2 days after birth. Selection pressure due to intensive use of antimicrobials in neonatal units (NNUs) has led to changes in the antimicrobial susceptibility patterns of organisms associated with nosocomial infections. Changes in the epidemiology of nosocomial infections and outbreaks of methicillin-resistant *Staphylococcus aureus*, ampicillin-resistant *E coli*, vancomycin-resistant enterococci and multi-resistant gram-negative organisms are increasingly being reported from NNUs.^{7–10} This is a major concern, especially since resistant gram-negative bacteria make the choice of empirical therapy more difficult and result in a poorer prognosis for the treated patient.¹¹

Guidance for health professionals on prescribing medicines to children in Britain is provided by the British National Formulary for Children (BNF-C).¹² However, a recent review of antibiotic policies in British and Irish NNUs found adherence to the BNF-C guidelines in only 69% of NNUs with widespread use of other antimicrobial regimes.¹³ In the present study, we used national laboratory surveillance data to determine whether the pathogens responsible for neonatal bacteraemia were sensitive to the antibiotic treatment recommended by the BNF-C and other commonly used antimicrobial regimes.

DESIGN

We extracted all reports of neonatal bacteraemia to the Health Protection Agency's (HPA) voluntary surveillance scheme in England and Wales from January 2006 until March 2008. Voluntary reporting to the HPA's LabBase2 surveillance database is continuous and captures routine microbiological information on bacteraemia from 90% of the hospital laboratories in England and Wales.¹⁴ A case of neonatal bacteraemia was defined as isolation of bacteria from blood culture from an infant younger than 29 days. Following the BNF-C's classification, early-onset infection was defined as presenting in neonates less than <48 h old and late-onset infection as occurring in neonates 2–28 days old. Bacteraemia reports included the neonate's date of birth, sex and hospital, the sample date, microorganism isolated, antimicrobial susceptibility and the reporting laboratory. Reports for the same strain from the same neonate within a 14-day period were removed. We grouped organisms according to whether they were gram-positive or gram-negative. Varying species of coagulase-negative staphylococci (CoNS) were combined, as were non-pyogenic streptococci, Enterobacteriaceae other than *E coli* (including *Klebsiella* spp., *Enterobacter* spp., *Morganella* spp., *Kluyvera* spp., *Citrobacter* spp., *Pantoea* spp., *Proteus* spp., *Salmonella* spp., *Serratia* spp. and *Yersinia enterocolitica*) and different species of *Enterococcus*, *Pseudomonas*, *Corynebacterium* and *Acinetobacter*. Pathogen groups were ranked according to frequency and we analysed their susceptibility to antimicrobials, or combinations of antimicrobials that are currently recommended for treating suspected neonatal septicaemia, based on antibiotic sensitivities reported by the laboratory. Organisms were regarded as susceptible if they were reported as being susceptible to either one or both of the antibiotics tested. Overall estimates of the proportion of pathogens susceptible to different treatment regimens are shown with and without CoNS, which rarely cause fulminant infections, are often due to

contamination and are usually treated based on antibiotic sensitivities. Microsoft Office Access 2003, Microsoft Office Excel 2003 and STATA v 10.1 were employed for data handling and analysis.

RESULTS

Reports of isolates from neonatal bacteraemia

Between January 2006 and March 2008, the surveillance scheme received 1516 reports of bacteria isolated from blood cultures taken from neonates <48 h old (early-onset) and 3482 reports for neonates 2–28 days old (late-onset). Fifty-one species of bacteria were reported for both age groups. The ranked distributions for the 20 most frequently reported organisms are presented for the two age groups in tables 1 and 2. The majority of bacteraemias were attributed to gram-positive organisms (82% in early-onset and 81% in late-onset bacteraemias). For early-onset bacteraemias, GBS (31%) were the most frequent pathogens, followed by CoNS (22%) and non-pyogenic streptococci (9%). *E coli* was the most frequently reported gram-negative organism (9%). For late-onset bacteraemias, CoNS accounted for the majority (45%) of isolates reported. Other common organisms were *S aureus* (13%) and *Enterococcus* spp. (7%). The most frequent gram-negative bacteria were Enterobacteriaceae other than *E coli* (9%).

Table 1 Bacteria isolated from blood cultures from neonates <48 h old, England and Wales, January 2006–March 2008

Organism	Number	% Of all isolates
Gram+	1246	82.2
GBS	477	31.5
CoNS	326	21.5
Non-pyogenic streptococci*	142	9.4
<i>Staphylococcus aureus</i>	75	4.9
<i>Enterococcus</i> spp.	49	3.2
<i>Micrococcus</i> sp.	35	2.3
<i>Streptococcus pneumoniae</i>	32	2.1
Diphtheroids	32	2.1
β-Haemolytic streptococci	28	1.8
<i>Listeria monocytogenes</i>	13	0.9
<i>Bacillus</i> sp.	10	0.7
Group A streptococci	5	0.3
<i>Propionibacterium</i> sp.	3	0.2
Other†	19	1.3
Gram–	270	17.8
<i>Escherichia coli</i>	137	9.0
Enterobacteriaceae‡	35	2.3
<i>Haemophilus influenzae</i>	34	2.2
<i>Pseudomonas</i> spp.	18	1.2
<i>Acinetobacter</i> spp.	12	0.8
<i>Haemophilus parainfluenzae</i>	8	0.5
<i>Haemophilus</i> sp.	4	0.3
Other§	22	1.5
Total	1516	100

**S viridans*, *S mitis*, *S oralis*, *S salivarius*, *S sanguinis* group, *S intermedius*, *S milleri*, *S anginosus*, *S acidominimus*, *S gordonii*, *Abiotrophia* spp., *Aerococcus* spp.

†*Streptococcus* group G, *Streptococcus* group C, *Streptococcus* group D, *Bacillus* other named, *Bacteroides* spp., *Lactococcus cremoris*, *Listeria* sp., *Streptococcus* group D, *Eubacterium* sp., *Gardnerella vaginalis*, *S anaerobic*, *S dysgalactiae*, *S equisimilis*.

‡*Klebsiella* spp., coliform, *Enterobacter* spp., *Morganella* spp., *Kluyvera* spp., *Citrobacter* spp., *Pantoea* sp., *Proteus* spp., *Salmonella paratyphi*, *Serratia* spp.

§*Moraxella* spp., *Stenotrophomonas maltophilia*, *Bacteroides* spp., *Neisseria* sp., *Sphingomonas paucimobilis*, *Aeromonas* spp., *Peptostreptococcus* spp., *Burkholderia cepaci*, *Oligella urethralis*, *Roseomonas* sp., *Weeksella virosa*. CoNS, coagulase-negative staphylococci; GBS, group B streptococcus.

Table 2 Bacteria isolated from blood cultures from neonates 2–28 days old, England and Wales, January 2006–March 2008

Organism	Number	% Of all isolates
Gram+	2811	80.7
CoNS	1569	45.1
<i>Staphylococcus aureus</i>	460	13.2
<i>Enterococcus</i> spp.	243	7.0
GBS	230	6.6
Non-pyogenic streptococci*	169	4.9
<i>Micrococcus</i> sp.	30	0.9
<i>Streptococcus pneumoniae</i>	25	0.7
<i>Bacillus</i> sp.	17	0.5
β-Haemolytic streptococci	17	0.5
Diphtheroids	13	0.4
Group A streptococci	8	0.2
Group D streptococci	7	0.2
<i>Listeria monocytogenes</i>	3	0.1
Other†	20	0.6
Gram–	671	19.3
Enterobacteriaceae‡	311	8.9
<i>Escherichia coli</i>	236	6.8
<i>Pseudomonas</i> spp.	66	1.9
<i>Acinetobacter</i> spp.	22	0.6
<i>Stenotrophomonas maltophilia</i>	9	0.3
<i>Neisseria meningitidis</i>	5	0.1
<i>Haemophilus influenzae</i>	4	0.1
Other§	18	0.5
Total	3482	100.0

**S viridans*, *S mitis*, *S salivarius*, *S oralis*, *S sanguinis* group, *S bovis*, *S milleri*, *S anginosus*, *S constellatus*, *S gordonii*, *S intermedius*, *Aerococcus* spp.

†*Lactococcus* spp., *Clostridium perfringens*, *Lactobacillus* sp., *Propionibacterium* sp., *Bacillus cereus*, *Gemella haemolysins*, *Stomatococcus mucilaginosus*, *Listeria* sp., *Micrococcus luteus*, *S anaerobic*, *Streptococcus* group G.

‡*Enterobacter* spp., *Klebsiella* spp., coliform, *Serratia* spp., *Citrobacter* spp., *Proteus* spp., *Morganella* spp., *Kluyvera* sp., *Salmonella munschaui*, *Yersinia enterocolitica*.

§*Moraxella* spp., *Neisseria* sp., *Haemophilus parainfluenzae*, *Ochrobactrum anthropi*, *Achromobacter* sp., *Alcaligenes faecalis*, *Bordetella bronchiseptica*, *Burkholderia cepacia*, *Haemophilus* sp., *Pasteurella multocida*, *Roseomonas* sp., *Sphingomonas paucimobilis*.

CoNS, coagulase-negative staphylococci; GBS, group B streptococcus.

Antimicrobial susceptibility of bacterial isolates

Among early-onset bacteraemias, the antimicrobial susceptibility of all isolates with known test results was 94% to the combination of penicillin+gentamicin, 100% to amoxicillin+cefotaxime, 98% to amoxicillin+penicillin and 96% to monotherapy with cefotaxime. After exclusion of CoNS results, susceptibility to the antibiotic regimes was the same or higher (table 3). The antimicrobial profiles for bacteria isolated from late-onset bacteraemias show that susceptibility was 69% to the combination flucloxacillin+gentamicin, 93% to amoxicillin+cefotaxime, 96% to amoxicillin+gentamicin and 78% to cefotaxime as a single agent. Susceptibility to the antimicrobial regimes was the same or higher after exclusion of results for CoNS (table 4).

DISCUSSION

We reviewed national laboratory surveillance data to guide empirical prescribing for neonatal septicemia. These data show that the aetiology of neonatal bacteraemia is diverse and varies by age, but the reported causative organisms are consistent with previous studies on pathogens causing bacteraemia in neonates.^{15–18} Gram-positive organisms caused the majority of neonatal bacteraemias. Only 18% of the reported early-onset bacteraemias were due to gram-negative organisms, with *E coli* being the most frequently reported. For late-onset bacteraemias, 19% of reports indicated a gram-negative

organism, with Enterobacteriaceae other than *E coli* most frequently reported.

The predominance of gram-negative infections in the neonatal care setting in the USA described in the very low-weight preterm infant population may reflect the implementation of universal GBS screening and use of IAP which has resulted in a significant decline in early-onset group B streptococcal infection in the USA.^{3 19} Routine screening for antenatal GBS carriage is not recommended in the UK and IAP use is considered only in the presence of known risk factors.^{20 21} The continued predominance of GBS as the main pathogen reported as early-onset infection in our study thus probably reflects the difference in IAP use between countries. As we have no data on birth weight/gestation we cannot provide a direct comparison with these US data. The decrease in GBS reported for late-onset bacteraemias compared to reports for early-onset bacteraemia demonstrates the shift from vertical transmission as the cause of neonatal sepsis during the first days of life to hospital-associated infections.^{22 23}

CoNS accounted for 22% of early-onset bacteraemia and nearly half of late-onset bacteraemia. CoNS are often regarded as contaminants, sometimes being excluded from analyses.^{3 16} However, one study found that more than half of late-onset CoNS-positive neonatal bacteraemias were considered to be true infections.²⁴ We were not able to determine the clinical significance of CoNS and other organisms of uncertain pathogenicity in our dataset due to lack of clinical parameters and have included these isolates in our overall analysis. However, as CoNS is predominantly a nosocomial pathogen, it is likely that the apparent early-onset CoNS bacteraemias predominantly reflect contamination rather than true infection. Even when isolated from late-onset bacteraemia, it is argued that CoNS is not a significant cause of mortality in neonates and does therefore not warrant specific empirical antibiotic therapy.²⁵ Presenting the analysis with and without CoNS allowed consideration of this viewpoint.

The overall susceptibility of blood culture isolates (excluding CoNS) to the antibiotics commonly used in British and Irish NNUs showed low resistance rates for most regimes. For early-onset bacteraemia, more than 94% of organisms were susceptible to regimens involving combinations of penicillin with either gentamicin or amoxicillin, amoxicillin combined with cefotaxime or cefotaxime monotherapy. For late-onset bacteraemia, more than 95% of organisms were susceptible to gentamicin with either flucloxacillin or amoxicillin and amoxicillin with cefotaxime, but only 79% were susceptible to cefotaxime monotherapy.

Despite an overall susceptibility of ≥93% for amoxicillin+cefotaxime, only 75% of the Enterobacteriaceae other than *E coli* and 46% of the *Pseudomonas* spp. from late-onset bacteraemias were susceptible to this combination. This is of particular concern since these pathogens are associated with significant morbidity and mortality in neonates and 11% of late-onset bacteraemias were due to these organisms in our study. Furthermore, cefotaxime is generally not considered to be effective clinically against enterococci, *Acinetobacter* spp. and *Listeria monocytogenes*, organisms which ranked among the 20 most commonly reported pathogens in our study.²⁶ Moreover, the frequent use of third-generation cephalosporins (eg, cefotaxime) has been shown to drive the development of resistant bacterial pathogens on neonatal intensive care units and has been linked to outbreaks caused by extended-spectrum β-lactamase-producing bacteria.^{27–29} de Man *et al* as well as other teams provided evidence that a regime avoiding

Table 3 Susceptibility to antimicrobials used for treatment of neonatal septicaemia in bacteria from blood culture taken from neonates less than 48 h old, England and Wales, January 2006–March 2008

Organism	Number	Pen+gent			Amox+ctx			Amox+pen			Ctx		
		Tested	(Of total) (%)	S (%)	Tested	(Of total) (%)	S (%)	Tested	(Of total) (%)	S (%)	Tested	(Of total) (%)	S (%)
Gram+	1246	927	(74)	93	426	(34)	100	752	(60)	97	92	(7)	96
GBS	477	371	(78)	99	247	(52)	100	402	(84)	100	56	(12)	100
CoNS	326	219	(67)	80	5	(2)	100	67	(21)	78	2	(1)	100
Non-pyogenic streptococci*	142	120	(85)	87	78	(55)	100	111	(78)	100	23	(16)	91
<i>Staphylococcus aureus</i>	75	60	(80)	100	1	(1)	100	11	(15)	73	0	(0)	NA
<i>Enterococcus</i> spp.	49	20	(41)	90	40	(82)	100	41	(84)	98	2	(4)	50
<i>Micrococcus</i> sp.	35	30	(86)	100	1	(3)	100	20	(57)	100	0	(0)	NA
<i>Streptococcus pneumoniae</i>	32	28	(88)	100	17	(53)	100	28	(88)	100	4	(13)	100
Other†	110	79	(72)	99	37	(34)	100	72	(65)	99	5	(5)	80
Gram–	270	177	(66)	99	164	(61)	99	114	(42)	99	104	(39)	97
<i>Escherichia coli</i>	137	119	(87)	98	98	(72)	100	69	(50)	99	66	(48)	100
Enterobacteriaceae‡	35	28	(80)	100	16	(46)	94	4	(11)	100	15	(43)	87
<i>Haemophilus influenzae</i>	34	1	(3)	100	23	(68)	100	19	(56)	100	11	(32)	100
Other§	64	29	(45)	100	27	(42)	96	22	(34)	100	12	(19)	92
Total excluding CoNS	1190	885	(74)	97	585	(49)	100	799	(67)	99	194	(16)	96
Total	1516	1104	(73)	94	590	(39)	100	866	(57)	98	196	(13)	96

**S. viridans*, *S. mitis*, *S. oralis*, *S. salivarius*, *S. sanguinis* group, *S. intermedius*, *S. milleri*, *S. anginosus*, *S. acidominimus*, *S. gordonii*, *Abiotrophia* spp., *Aerococcus* spp.

†Diphtheroids, β-haemolytic streptococci, *Listeria monocytogenes*, *Bacillus* sp., *Streptococcus* group A, *Propionibacterium* sp., *Streptococcus* group G, *Streptococcus* group C, *Streptococcus* group D, *Bacillus* other named, *Bacteroides* spp., *Lactococcus cremoris*, *Listeria* sp., *Streptococcus* group D, *Eubacterium* sp., *Gardnerella vaginalis*, *S. anaerobic*, *S. dysgalactiae*, *S. equisimilis*.

‡*Klebsiella* spp., coliform, *Enterobacter* spp., *Morganella* spp., *Kluyvera* sp., *Citrobacter* spp., *Pantoea* sp., *Proteus* spp., *Salmonella paratyphi*, *Serratia* spp.

§*Pseudomonas* spp., *Acinetobacter* spp., *Haemophilus parainfluenzae*, *Haemophilus* sp., *Moraxella* spp., *Stenotrophomonas maltophilia*, *Bacteroides* spp., *Neisseria* sp., *Sphingomonas paucimobilis*, *Aeromonas* spp., *Peptostreptococcus* spp., *Burkholderia cepaci*, *Oligella urethralis*, *Roseomonas* sp., *Weeksella virosa*.

Amox, amoxicillin; CoNS, coagulase-negative staphylococci; ctx, cefotaxime; GBS, group B streptococcus; gent, gentamicin; Pen, penicillin; S, susceptibility.

Table 4 Susceptibility to antimicrobials used for treatment of neonatal septicaemia in bacteria from blood culture taken from neonates 2–28 days old, England and Wales, January 2006–March 2008

Organism	Number	Flucloxac+gent			Amox+ctx			Amox+gent			Ctx		
		Tested	(Of total) (%)	S (%)	Tested	(Of total) (%)	S	Tested	(Of total) (%)	S (%)	Tested	(Of total) (%)	S (%)
Gram+	2811	1721	(61)	62	443	(16)	98	1143	(41)	97	115	(4)	79
CoNS	1569	1052	(67)	41	33	(2)	91	404	(26)	92	6	(<1)	17
<i>Staphylococcus aureus</i>	460	425	(92)	98	25	(5)	96	320	(70)	100	13	(3)	92
<i>Enterococcus</i> spp.	243	104	(43)	83	156	(64)	99	174	(72)	99	19	(8)	37
GBS	230	35	(15)	100	117	(51)	100	105	(46)	100	36	(16)	100
Non-pyogenic streptococci*	169	40	(24)	95	70	(41)	100	69	(41)	100	20	(12)	95
<i>Micrococcus</i> sp.	30	24	(80)	100	0	(0)	NA	22	(73)	100	0	(0)	NA
<i>Streptococcus pneumoniae</i>	25	9	(36)	100	15	(60)	100	10	(40)	100	7	(28)	100
Other†	85	32	(38)	81	27	(32)	81	39	(46)	100	14	(16)	64
Gram–	671	481	(72)	96	304	(45)	85	506	(75)	93	250	(37)	77
Enterobacteriaceae‡	311	224	(72)	94	122	(39)	75	222	(71)	95	122	(39)	69
<i>Escherichia coli</i>	236	181	(77)	98	143	(61)	98	195	(83)	93	104	(44)	95
<i>Pseudomonas</i> spp.	66	54	(82)	94	13	(20)	46	55	(83)	95	9	(14)	11
Other§	58	22	(38)	95	26	(45)	96	34	(59)	82	15	(26)	60
Total excluding CoNS	1913	1150	(60)	95	714	(37)	93	1245	(65)	97	359	(19)	79
Total	3482	2202	(63)	69	747	(21)	93	1649	(47)	96	365	(10)	78

**S. viridans*, *S. mitis*, *S. salivarius*, *S. oralis*, *S. sanguinis* group, *S. bovis*, *S. milleri*, *S. anginosus*, *S. constellatus*, *S. gordonii*, *S. intermedius*, *Aerococcus* spp.

†*Bacillus* sp., β-haemolytic streptococci, diphtheroids, *Streptococcus* group A, *Streptococcus* group D, *Listeria monocytogenes*, *Lactococcus* spp., *Clostridium perfringens*, *Lactobacillus* sp., *Propionibacterium* sp., *Bacillus cereus*, *Gemella haemolysins*, *Stomatococcus mucilaginosus*, *Listeria* sp., *Micrococcus luteus*, *S. anaerobic*, *Streptococcus* group G.

‡*Enterobacter* spp., *Klebsiella* spp., coliform, *Serratia* spp., *Citrobacter* spp., *Proteus* spp., *Morganella* spp., *Kluyvera* sp., *Salmonella munschaui*, *Yersinia enterocolitica*.

§*Acinetobacter* spp., *Stenotrophomonas maltophilia*, *Neisseria meningitidis*, *Haemophilus influenzae*, *Moraxella* spp., *Neisseria* sp., *Haemophilus parainfluenzae*, *Ochrobactrum anthropi*, *Achromobacter* sp., *Alcaligenes faecalis*, *Bordetella bronchiseptica*, *Burkholderia cepacia*, *Haemophilus* sp., *Pasteurella multocida*, *Roseomonas* sp., *Sphingomonas paucimobilis*.

Amox, amoxicillin; CoNS, coagulase-negative staphylococci; ctx, cefotaxime; Flucloxac, flucloxacillin; GBS, group B streptococcus; gent, gentamicin; Pen, penicillin; S, susceptibility.

cefotaxime decreases the incidence of resistant gram-negative bacteria on NNUs.^{28 30} Minimising antimicrobial resistance is particularly important in the NNU setting where the newborn patients have a low grade of endogenous flora and will inevitably be colonised by nosocomial strains.

There were two limitations to our study. Due to the lack of clinical parameters we were not able to determine whether isolation of organisms from blood reflected true infection or contamination. Also, it was not possible to correlate in vitro susceptibility testing with in vivo clinical effectiveness. Future

work aimed at addressing these issues might include linkage of LabBase2 data to clinical and prescribing data. Additionally, the low frequency of susceptibility tests for some antimicrobial/organism combinations limited the interpretations of results. In particular, cefotaxime susceptibility was reported for only 4–7% of gram-positive organisms and for 37–39% of gram-negative organisms. This is probably due to laboratories choosing their panel of antibiotics for susceptibility testing based on the initial identification of the organism rather than the guidance for empirical prescribing. For example, cefotaxime is not included in national surveillance of the antibiotic susceptibility of staphylococci, reflecting the fact that it is not routinely tested.³¹

This analysis of national laboratory surveillance data has shown that current guidelines for empirical antimicrobial therapy in neonates with sepsis appear appropriate, given the results from susceptibility testing of causative organisms. However, cefotaxime-based treatments should not be recommended for empirical treatment of suspected neonatal sepsis in the UK, because of the lower levels of susceptibility in important causative organisms and the need to avoid exerting selective pressure particularly on gram-negative bacteria. We would suggest that NNUs switch to gentamicin-based regimens rather than cefotaxime-based regimens. More robust surveillance data, linked to clinical and treatment data, could further inform rational antibiotic prescribing based on microbiological findings as well as the type of patients affected.

Acknowledgements The authors thank the microbiology laboratories across the country for their considerable efforts in reporting to LabBase2. The authors also thank our colleagues in the iCAP Group (Improving Antibiotic Prescribing in Primary care): Professor Ian Wong, The School of Pharmacy; Dr Sonia Saxena, Imperial College London; Dr Paul Long, The School of Pharmacy; Dr Alessandro Porta, St George's Hospital; Helen Bird, St George's Hospital; Ruthie Birger, Imperial College London; Saidia Mehonic; Shama Wagle, BNF Publications; Joanna Murray, Imperial College London; Elizabeth Koshy, Imperial College London; Yingfen Hsia, The School of Pharmacy. We also thank Theresa Lamagni for her helpful advice.

Competing interests None.

Provenance and peer review Not commissioned; externally peer reviewed.

REFERENCES

1. **Modi N**, Doré CJ, Saraswatula A, *et al*. A case definition for national and international neonatal bloodstream infection surveillance. *Arch Dis Child Fetal Neonatal Ed* 2009;**94**:F8–12.
2. **McDuffie RS Jr**, McGregor JA, Gibbs RS. Adverse perinatal outcome and resistant Enterobacteriaceae after antibiotic usage for premature rupture of the membranes and group B streptococcus carriage. *Obstet Gynecol* 1993;**82**:487–9.
3. **Stoll BJ**, Hansen NI, Higgins RD, *et al*. Very low birth weight preterm infants with early onset neonatal sepsis: the predominance of gram-negative infections continues in the National Institute of Child Health and Human Development Neonatal Research Network, 2002–2003. *Pediatr Infect Dis J* 2005;**24**:635–9.
4. **Nambiar S**, Singh N. Change in epidemiology of health care-associated infections in a neonatal intensive care unit. *Pediatr Infect Dis J* 2002;**21**:839–42.
5. **Gilbert R**. Prenatal screening for group B streptococcal infection: gaps in the evidence. *Int J Epidemiol* 2004;**33**:2–8.
6. **Heath PT**, Balfour G, Weisner AM, *et al*. Group B streptococcal disease in UK and Irish infants younger than 90 days. *Lancet* 2004;**363**:292–4.
7. **Musoke RN**, Revathi G. Emergence of multidrug-resistant gram-negative organisms in a neonatal unit and the therapeutic implications. *J Trop Pediatr* 2000;**46**:86–91.
8. **Nambiar S**, Herwaldt LA, Singh N. Outbreak of invasive disease caused by methicillin-resistant *Staphylococcus aureus* in neonates and prevalence in the neonatal intensive care unit. *Pediatr Crit Care Med* 2003;**4**:220–6.
9. **Sherer CR**, Sprague BM, Campos JM, *et al*. Characterizing vancomycin-resistant enterococci in neonatal intensive care. *Emerging Infect Dis* 2005;**11**:1470–2.
10. **Toltzis P**. Antibiotic-resistant gram-negative bacteria in hospitalized children. *Clin Lab Med* 2004;**24**:363–80.
11. **Kang CI**, Kim SH, Park WB, *et al*. Bloodstream infections caused by antibiotic-resistant gram-negative bacilli: risk factors for mortality and impact of inappropriate initial antimicrobial therapy on outcome. *Antimicrob Agents Chemother* 2005;**49**:760–6.
12. **The Paediatric Formulary Committee**. BNF for children. British Medical Association, Royal Pharmaceutical Society of Great Britain, Royal College of Paediatrics and Child Health, Neonatal and Paediatric Pharmacists Group, 2008.
13. **Fernando AM**, Heath PT, Menson EN. Antimicrobial policies in the neonatal units of the United Kingdom and Republic of Ireland. *J Antimicrob Chemother* 2008;**61**:743–5.
14. **Reacher MH**, Shah A, Livermore DM, *et al*. Bacteraemia and antibiotic resistance of its pathogens reported in England and Wales between 1990 and 1998: trend analysis. *BMJ* 2000;**320**:213–16.
15. **Berger A**, Salzer HR, Weninger M, *et al*. Septicaemia in an Austrian neonatal intensive care unit: a 7-year analysis. *Acta Paediatr* 1998;**87**:1066–9.
16. **Hyde TB**, Hilger TM, Reingold A, *et al*. Trends in incidence and antimicrobial resistance of early-onset sepsis: population-based surveillance in San Francisco and Atlanta. *Pediatrics* 2002;**110**:690–5.
17. **McGuire W**, Clerihew L, Fowlie PW. Infection in the preterm infant. *BMJ* 2004;**329**:1277–80.
18. **Sharland M**. The use of antibacterials in children: a report of the Specialist Advisory Committee on Antimicrobial Resistance (SACAR) Paediatric Subgroup. *J Antimicrob Chemother* 2007;**60**(Suppl 1):i15–26.
19. **CDC**. Prevention of perinatal group B streptococcal disease – revised guidelines from CDC. *MMWR Morb Mortal Wkly Rep* 2002;**51**:1–22.
20. **Colbourn TE**, Asseburg C, Bojke L, *et al*. Preventive strategies for group B streptococcal and other bacterial infections in early infancy: cost effectiveness and value of information analyses. *BMJ* 2007;**335**:655.
21. **RCOG Royal College of Obstetricians and Gynaecologists**. Prevention of early onset neonatal group B streptococcal disease – guideline no. 36. Guidelines and Audit Committee of the RCOG. 1 November 2003, London.
22. **Lautenbach E**, Polk RE. Resistant gram-negative bacilli: a neglected healthcare crisis? *Am J Health Syst Pharm* 2007;**64**:S3–21; quiz S22–4.
23. **Orsi GB**, d'Ettorre G, Panero A, *et al*. Hospital-acquired infection surveillance in a neonatal intensive care unit. *Am J Infect Control* 2009;**37**:201–3.
24. **Huang SY**, Tang RB, Chen SJ, *et al*. Coagulase-negative staphylococcal bacteremia in critically ill children: risk factors and antimicrobial susceptibility. *J Microbiol Immunol Infect* 2003;**36**:51–5.
25. **Isaacs D**. A ten year, multicentre study of coagulase negative staphylococcal infections in Australasian neonatal units. *Arch Dis Child Fetal Neonatal Ed* 2003;**88**:F89–93.
26. **Gilbert DN**, Moellering RC, Eliopoulos GM, *et al*. Comparison of antimicrobial spectra. In: *The Sanford guide to antimicrobial therapy* 2008. Sperryville, Virginia, USA: Antimicrobial Therapy, 2008:65–70.
27. **Bryan CS**, John JF Jr, Pai MS, *et al*. Gentamicin vs cefotaxime for therapy of neonatal sepsis. Relationship to drug resistance. *Am J Dis Child* 1985;**139**:1086–9.
28. **de Man P**, Verhoeven BA, Verbrugh HA, *et al*. An antibiotic policy to prevent emergence of resistant bacilli. *Lancet* 2000;**355**:973–8.
29. **Modi N**, Damjanovic V, Cooke RW. Outbreak of cephalosporin resistant *Enterobacter cloacae* infection in a neonatal intensive care unit. *Arch Dis Child* 1987;**62**:148–51.
30. **Le J**, Nguyen T, Okamoto M, *et al*. Impact of empiric antibiotic use on development of infections caused by extended-spectrum beta-lactamase bacteria in a neonatal intensive care unit. *Pediatr Infect Dis J* 2008;**27**:314–18.
31. **Hope R**, Livermore DM, Brick G, *et al*. Non-susceptibility trends among staphylococci from bacteraemias in the UK and Ireland, 2001–06. *J Antimicrob Chemother* 2008;**62**(Suppl 2):ii65–74.